

Tensor sectors retained by the original operators

$$A_p \ell_{ab} = \alpha_{ab} \ell_{ab}, \quad \alpha_{ab} = (2a - k)\delta \log p + i(2b - k)\gamma \log p$$

ℓ_{ab}
 α_{ab}

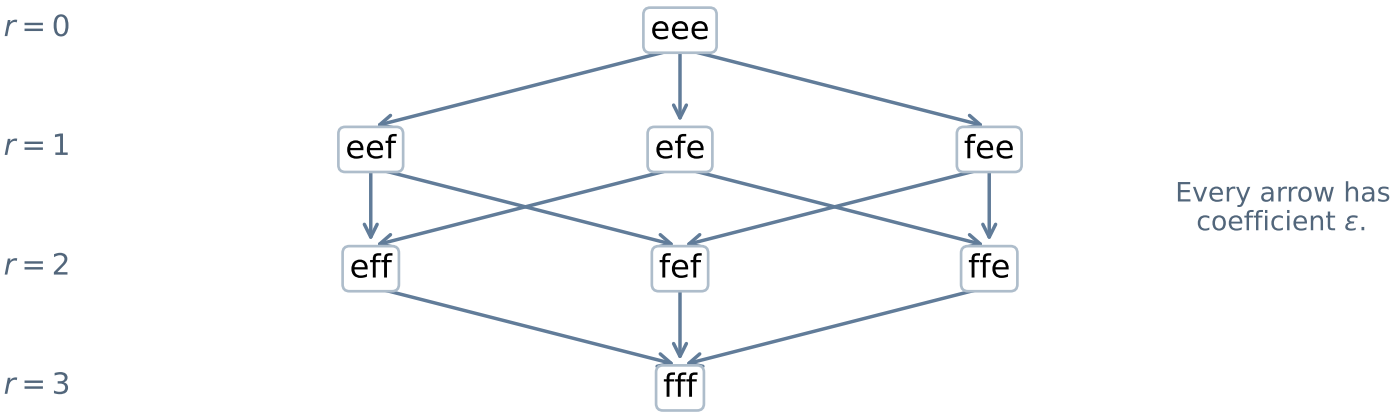
$\ell_{k-a, k-b}$
 $-\alpha_{ab}$

$A_p^{[2]}(\ell_{ab} \otimes \ell_{k-a, k-b}) = 0$

$q = (k + 1)^2$ independent vectors; uncentred exponent $k \log p$ is retained.

Nilpotent tensor block: $Re = \varepsilon f, Rf = 0, \varepsilon > 0$

$$X_3 = R \otimes I \otimes I + I \otimes R \otimes I + I \otimes I \otimes R$$



$$\ker X_3 = \text{span}\{ffe - fef, fef - eff, fff\}$$

Words denote the displayed tensor factors. The original orthonormal e, f and their tensor metric are retained; the three kernel vectors need not be orthogonal.

$$\dim \ker R^{[t]} = \sum_{j=0}^t \binom{t}{j} (q - 2)^{t-j} \binom{j}{\lfloor j/2 \rfloor}$$

Complete receiver proofs: OLR23–31. Lattice lifts: OLR16–17.
 Source locality: Connes–Consani, arXiv:2606.06604v1, Definition defnloc.
 These local tensor sectors do not impose a purity theorem on the original object.